24/11/2020

Theorem: Suppose G is a system with a state-space realization

$$G := \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ C_2 & D_{21} & 0 \end{array}\right]$$

for the T.M to be in H_2

to ensure well-posedness (this is a sufficiency cond^n)

and ① (A, B_2) is stabilizable and (A, C_2) is detectable

② $\forall\, \omega \in \mathbb{R}$, $\operatorname{rank}\begin{bmatrix} A-j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix} = \#$ of columns

$\operatorname{rank}\begin{bmatrix} A-j\omega I & B_1 \\ C_2 & D_{21} \end{bmatrix} = \#$ of rows

③ $D_{12}^* D_{12} > 0$ & $D_{21} D_{21}^* > 0$

Then the optimal stabilizing controller for the H_2-synthesis problem is

$$\mathscr{K} \quad K = \left[\begin{array}{c|c} (A + B_2 F + L C_2) & -L \\ \hline F & 0 \end{array}\right]$$

$\dot{x}_K = A_K x_K + B_K y$
$u = C_K x_K + D_K y$

where $F = -(D_{12}^* D_{12})^{-1}(B_2^* X + D_{12}^T C_1)$, $\quad L = -(Y C_2^* + B_1 D_{21}^*)(D_{21} D_{21}^*)^{-1}$

with $X = \operatorname{Ric}(H)$ and $Y = \operatorname{Ric}(J)$

$$H = \begin{bmatrix} A - B_2 (D_{12}^* D_{12})^{-1} D_{12}^* C_1 & -B_2 (D_{12}^* D_{12})^{-1} B_2^* \\ -C_1^* \left(I - D_{12} (D_{12}^* D_{12})^{-1} D_{12}^*\right) C_1 & -\left(A - B_2 (D_{12}^* D_{12})^{-1} D_{12}^* C_1\right)^* \end{bmatrix}$$

$$J = \begin{bmatrix} \left(A - B_1 D_{21}^* (D_{21}^* D_{21})^{-1} C_2\right)^* & -C_2^* (D_{21} D_{21}^*)^{-1} C_2 \\ -B_1 \left(I - D_{21}^* (D_{21} D_{21}^*)^{-1} D_{21}\right) B_1^* & -\left(A - B_1 D_{21}^* (D_{21} D_{21}^*)^{-1} C_2\right) \end{bmatrix}$$

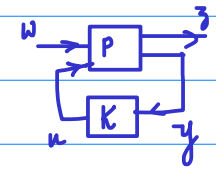
Controller/Observer model:

$$G_c := \begin{bmatrix} A + B_2 F & I \\ C_1 + D_{12} F & 0 \end{bmatrix} \qquad G_f = \begin{bmatrix} A + L C_2 & B_1 + L D_{21} \\ I & 0 \end{bmatrix}$$

Minimum H_2-norm achieved:

$$\| G_c B_1 \|_2^2 + \| (D_{12}^* D_{12})^{1/2} F G_f \|^2 = \text{trace}\left[B_1^* X B_1\right] + \text{trace}\left[D_{12}^* D_{12} F Y F^*\right]$$

Proposition: A necessary and sufficient condition for the existence of an internally stabilizing K for the system in figure is that (C_2, A, B_2) is stabilizable & detectable.



$$\boxed{\begin{aligned} \dot{x} &= Ax + B_1 w + B_2 u \\ z &= C_1 x + D_{11} w + D_{12} u \\ y &= C_2 x + D_{21} w + D_{22} u \end{aligned}}$$

In that case, one such controller is given by —

$$\longrightarrow \quad K(s) = \left[\begin{array}{c|c} A + B_2 F + L C_2 + L D_{22} F & -L \\ \hline F & 0 \end{array}\right]$$

where F & L are matrices such that $(A + B_2 F)$ & $(A + L C_2)$ are Hurwitz.

State-feedback H_2 synthesis:

$y = x$ (State-feedback)

$$P = \begin{array}{c} \\ x \\ z \\ y \end{array} \begin{array}{c} \begin{array}{cccc} x & z & w & u \end{array} \\ \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ I & 0 & 0 \end{array}\right] \end{array}$$

$\dot{x}_K = A_K x_K + B_K y$

$u = D_K y \Rightarrow u = D_K x$

$K = D_K$?

Theorem: There exists a feedback gain $K(s) = D_K$ that internally stabilizes P and satisfies $\| F_\ell(P, K) \|_2 < 1$

if and only if $\exists\, X \in \mathbb{R}^{n \times n}$, $Z \in \mathbb{R}^{q \times q}$ and $Y \in \mathbb{R}^{m \times n}$ s.t.

$$D_K = Y X^{-1}$$

and inequalities:

$$\begin{bmatrix} A & B_2 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} + \begin{bmatrix} X & Y^* \end{bmatrix} \begin{bmatrix} A^* \\ B_2^* \end{bmatrix} + B_1 B_1^* < 0$$

$$\begin{bmatrix} X & (C_1 X + D_{12} Y)^* \\ (C_1 X + D_{12} Y) & Z \end{bmatrix} > 0$$

$$\text{Trace}(Z) < 1$$

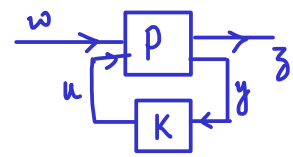
are satisfied.

LMI (Linear matrix inequality)

sedumi ← (Matlab)

$\boxed{\text{LMI solvers}}$ ←

H_∞ - control problem:



Find a controller $K(s)$ such that

(I) The closed loop system is internally stable.
(II) it minimizes $\|\mathcal{F}_\ell(P,K)\|_{H_\infty}$

} Hard problem

$$\|\mathcal{F}_\ell(P,K)\|_{H_\infty} := \sup_{\substack{w(t)\in \mathcal{L}_2 \\ \|w(t)\|\neq 0}} \frac{\|z\|_2}{\|w\|_2}$$

$\boxed{\text{worst-case effect of noise on } w}$

* Suboptimal H_∞-problem

Given $\gamma > 0$, find $K(s)$ (if it exists) such that

(i) the closed loop system is internally stabilized.
(ii) $\|\mathcal{F}_\ell(P,K)\|_{H_\infty} < \gamma$.

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$$G := \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ C_2 & D_{21} & 0 \end{array}\right]$$

and (1) (A, B_1) is stabilizable and (A, C_1) is detectable
(2) (A, B_2) is stabilizable and (A, C_2) is detectable
(3) $D_{12}^T [C_1 \;\; D_{12}] = [0 \;\; I]$
(4) $D_{21} [B_1^T \;\; D_{21}^T] = [0 \;\; I]$

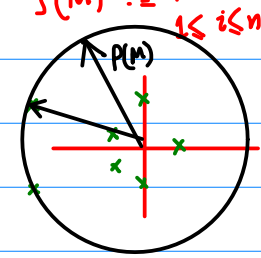
Then $\exists\, K$ that internally stabilizes the system and $\|\mathcal{F}_\ell(P,K)\|_\infty < \gamma$ if and only if

✓ (i) $N \in \text{domain}(Ric)$ and $X = Ric(\underline{N}) \geq 0$
✓ (ii) $J \in \text{domain}(Ric)$ and $Y = Ric(J) \geq 0$
(iii) $\rho(\underline{X}Y) < \gamma^2$

}

Spectral radius

$$\rho(M) := \max_{1\leq i\leq n} |\lambda_i(M)|$$



$$N := \begin{bmatrix} A & 1/\gamma^2 B_1 B_1^* - B_2 B_2^* \\ -C_1^* C_1 & -A^* \end{bmatrix} \qquad J := \begin{bmatrix} A^* & \frac{1}{\gamma^2} C_1^* C_1 - C_2^* C_2 \\ -B_1 B_1^* & -A \end{bmatrix}$$

where $\gamma > 0$

When the above conditions are satisfied, then one such controller

$$K_{sub}(s) = \left[\begin{array}{c|c} A_{closed} & -ZL \\ \hline F & 0 \end{array}\right]$$

$$A_{closed} := A + \frac{1}{\gamma^2} B_1 B_1^* X + B_2 F + ZLC_2 \,, \quad F := -B_2^* X \,, \quad L = -YC_2^*$$

$$Z := \left(I - \frac{1}{\gamma^2} YX\right)^{-1}$$